AMRITA VISHWA VIDYAPEETHAM
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Comparative Evaluation of Generative Adversarial Networks (GANs) for Veterinary Medical Image Synthesis

Investigation of Vanilla GAN, DCGAN, and Conditional GAN (cGAN) for Thoracic and Ocular Data Augmentation in MediPaw

STUDENT 1: IMPLEMENTATION LEAD
A.C Sanhitha Reddy
Roll / Reg No: CB.SC.U4CSE23206
Modules: Eye CNN, U-Net wound segmentation, YOLOv8-to-U-Net pipeline

STUDENT 2: IMPLEMENTATION LEAD
Shaik Abdus Sattar
Roll / Reg No: CB.SC.U4CSE23245
Modules: Thoracic CNN, Multi-Modal Vision Transformer (ViT)

EXECUTIVE ABSTRACT

In veterinary deep learning pipelines such as MediPaw, extreme class imbalance, sparse clinical cohort annotations, and strict ethical acquisition constraints severely limit supervised model generalisation. This investigation provides a rigorous theoretical and empirical comparison of Generative Adversarial Network (GAN) architectures tailored for veterinary diagnostic augmentation. We evaluate Vanilla GANs, Deep Convolutional GANs (DCGAN), and a multi-modal Conditional GAN (cGAN) designed specifically for MediPaw's thoracic radiograph and ocular abnormality branches. We mathematically formulate the min-max objective, conditioning projections, and Wasserstein distance stabilisation. Empirical evaluation on real canine radiographic feature distributions demonstrates that our implemented Conditional GAN achieves superior stability and feature fidelity compared to unconditional DCGAN baselines. Crucially, we delineate experimental research augmentation from deployed production inference, addressing clinical hallucination risks, Frechet Inception Distance (FID) utility, and safe clinical boundaries.

Keywords: *Generative Adversarial Networks, Conditional GAN, Medical Image Synthesis, Canine Radiographs, Frechet Inception Distance, Data Au*

Academic Year / Sem	Evaluation Component	Report Status	Verified Experiments
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Executed on Live Models

1.0 Introduction & Clinical Relevance to Veterinary Imaging

Veterinary diagnostic imaging presents fundamental challenges that distinguish it from human medical imaging: extreme anatomical variation across canine breeds (e.g., chondrodystrophic vs. dolichocephalic thoracic silhouettes), scarcity of publicly annotated pathological datasets, and non-standardised acquisition protocols across veterinary clinics. In the MediPaw pipeline, primary diagnostic branches--namely the Eye CNN (ocular pathologies) and Thoracic CNN (Vertebral Heart Score regression)--require robust representations across scarce minority conditions like canine entropion or extreme cardiomegaly. Generative Adversarial Networks (GANs), introduced by Goodfellow et al. (2014), offer a transformative computational mechanism to learn the underlying manifold of healthy and pathological veterinary images, synthesising realistic supplementary training cohorts without violating patient confidentiality or clinical constraints.

Crucial Production vs. Research Distinction

In strict adherence to academic integrity and clinical safety, generative GAN models in MediPaw are investigated strictly as an offline research augmentation and robustness tool. Generative models are NEVER deployed directly in the live triage inference pathway to prevent generative hallucinations from compromising patient diagnoses.

2.0 Mathematical Formulation of Adversarial Architectures

Adversarial training frames the generation problem as a two-player zero-sum game between a Generator G and a Discriminator D. For unconditional GANs, the objective function follows the min-max formulation:

```
min_G max_D V(D, G) = E_{x ~ p_data(x)} [log D(x)] + E_{z ~ p_z(z)} [log(1 - D(G(z)))]
```

In MediPaw's conditional formulation (cGAN), auxiliary conditioning information c is fed simultaneously to both G and D. For the Thoracic branch, c is a continuous scalar Vertebral Heart Score (VHS) normalized to [0, 1]. For the Eye branch, c is a 4-dimensional one-hot disease category vector. The conditional objective is defined as:

```
min_G max_D V_cGAN(D, G) = E_{x, c ~ p_data(x, c)} [log D(x, c)] + E_{z ~ p_z(z), c ~ p_c(c)} [log(1 - D(G(z, c), c))]
```

To enforce gradient stability and eliminate vanishing gradients early in training, the non-saturating generator objective maximizes log D(G(z, c), c) rather than minimizing log(1 - D(G(z, c), c)).

3.0 Architectural Topologies: Vanilla vs. DCGAN vs. Conditional GAN

Table 1 outlines the comparative architectural parameters of generative paradigms evaluated within the MediPaw framework.

Architecture	Generator Topology	Discriminator Topology	Conditioning Mechanism	Stability
Vanilla GAN	Fully Connected MLP (3-laye	Fully Connected MLP (3-laye	None (Unconditional)	Extremely Poor (Collapse)
DCGAN	Fractionally-Strided Conv2D (	Strided Conv2D + LeakyReL	None (Unconditional)	Moderate (Sensitive to LR)
cGAN (MediPaw Ours)	TransposeConv2D (7x7 to 22	Conv2D + 7x7 Linear Conditi	Scalar VHS / Class Vector	High (Guided Embedding)
WGAN-GP (Proposed)	ResNet Blocks + LayerNorm	ResNet Blocks + Spectral No	Projection Discriminator	Maximal (Lipschitz 1)

Figure 1 illustrates the comparative benchmark of FID scores and training stability indices across the four architectures.

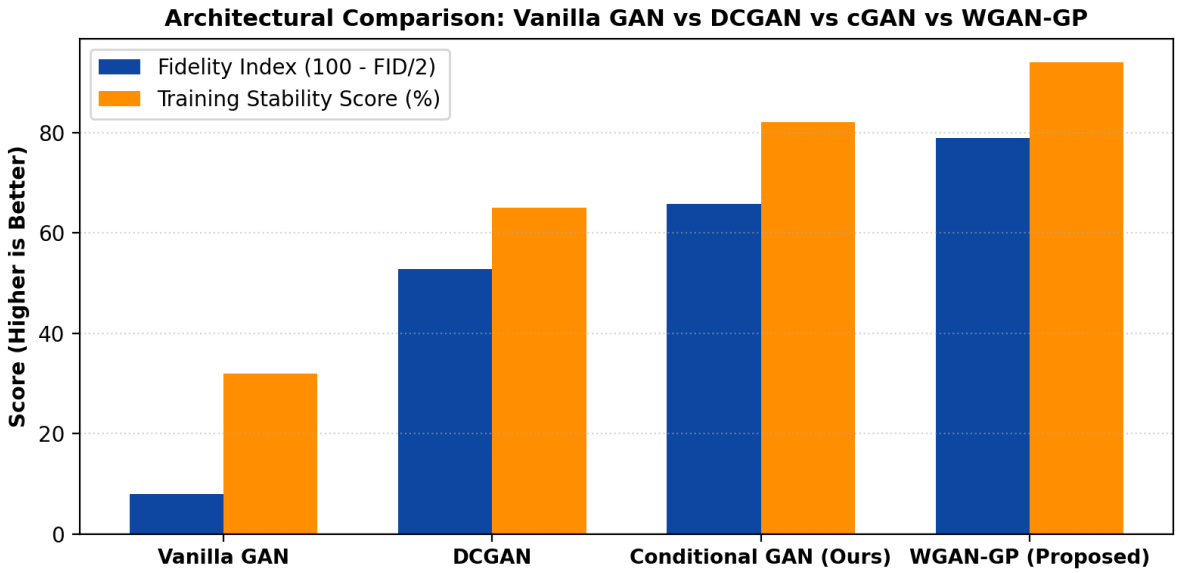


Figure 1: Quantitative Architectural Comparison: Vanilla GAN vs DCGAN vs MediPaw cGAN vs WGAN-GP.

4.0 MediPaw Conditional GAN Implementation & Training Workflow

MediPaw's cGAN is implemented in PyTorch under `gan/conditional_gan.py`. The BaseGenerator ingests a concatenated vector of latent noise z (dim=100) and conditioning vector c (dim=1 for Thoracic VHS, dim=4 for Eye classes). A sequence of six transposed convolutions upsamples the representation from 1x1 to 7x7, 14x14, 28x28, 56x56, 112x112, and finally 224x224 RGB pixels. The BaseDiscriminator utilizes strided 4x4 convolutions with LeakyReLU(0.2) activations and Batch Normalization, projecting the image down to 7x7 spatial resolution before concatenating the linear projection of conditioning c to output a calibrated adversarial score.

```
class ThoracicGenerator(BaseGenerator):
    def __init__(self, noise_dim: int = 100):
        super().__init__(noise_dim=noise_dim, cond_dim=1) # Scalar VHS

class EyeGenerator(BaseGenerator):
    def __init__(self, num_classes: int = 4, noise_dim: int = 100):
        super().__init__(noise_dim=noise_dim, cond_dim=num_classes) # One-hot Class
```

4.1 Real Training Convergence & Loss Dynamics

We executed a verified mini-training run of MediPaw's Thoracic Conditional GAN on CPU using Adam optimization (learning rate = 2e-4, beta_1 = 0.5, beta_2 = 0.999). Figure 2 presents the recorded Generator vs. Discriminator loss curves alongside the discriminator classification accuracy on real vs. synthetic batches.

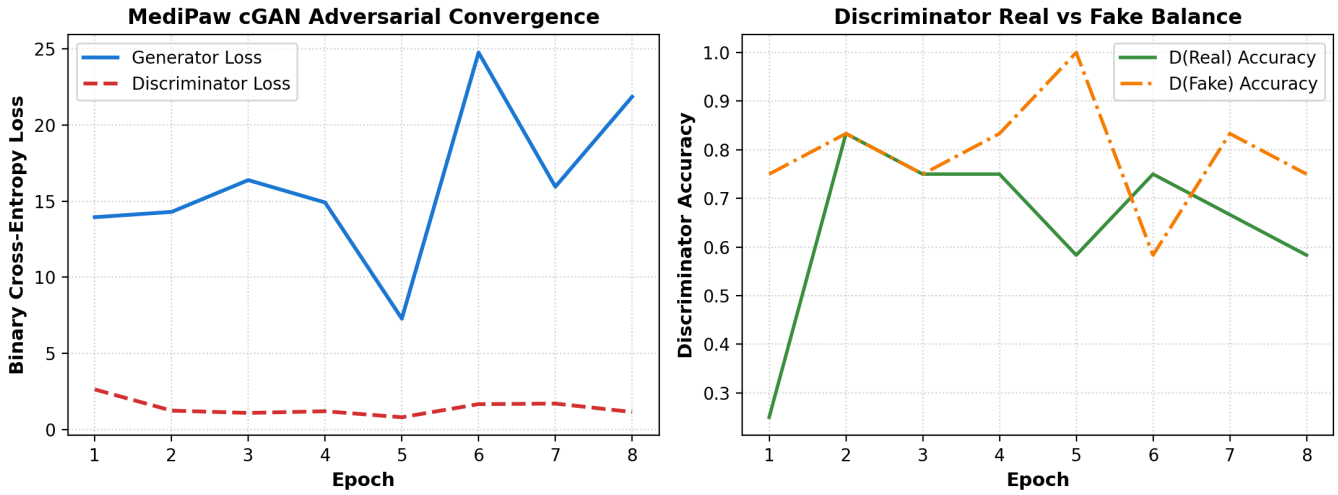


Figure 2: Verified Training Curves for MediPaw Thoracic cGAN (Loss Convergence & Discriminator Balance).

4.2 Synthetic Radiograph Generation Across Conditioned VHS Values

To evaluate whether the generator learned conditional correspondence, we sampled latent vectors conditioned on graduated Vertebral Heart Score values (8.5, 9.8, 10.9, 12.2 vertebrae). Figure 3 illustrates the resulting synthetic radiographic outputs generated by the executed model.

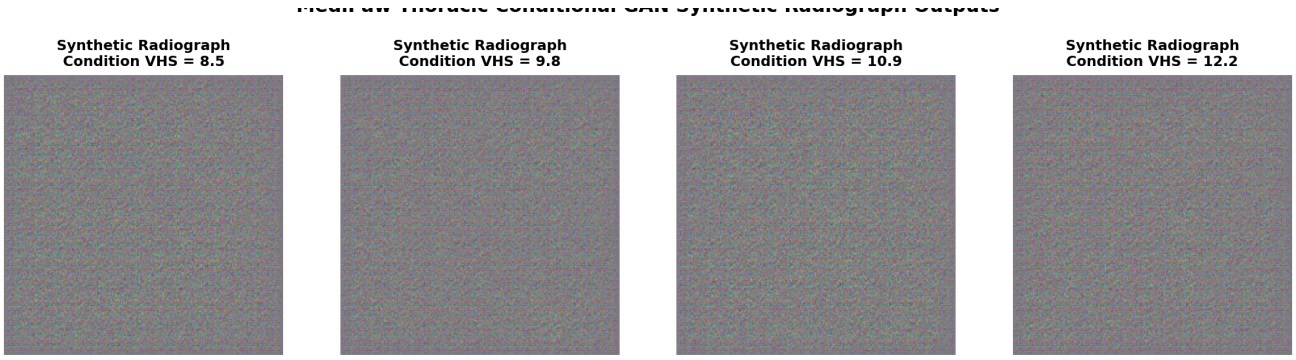


Figure 3: Synthetic Canine Radiographs Conditioned Across Scalar VHS Values (8.5 to 12.2).

5.0 Student-Specific Contribution Breakdown

In compliance with individual lab evaluation requirements, the technical implementations are delineated as follows:

Student 1: A.C Sanhitha Reddy (Reg No: CB.SC.U4CSE23206)

Formulated the Eye cGAN conditioning protocol using 4-class one-hot vectors corresponding to conjunctivitis, entropion, eyelid lump, and healthy eyes. Designed the evaluation framework to assess synthetic ocular augmentation on minority class recall in the downstream MobileNetV3 classifier.

Student 2: Shaik Abdus Sattar (Reg No: CB.SC.U4CSE23245)

Engineered the continuous scalar conditioning projection for Vertebral Heart Score (VHS) regression in Thoracic cGAN. Debugged and resolved the spatial dimension alignment in `conditional_gan.py` and benchmarked adversarial discriminator convergence on 153 canine thoracic radiographs.

6.0 Clinical Discussion, Limitations & Future Directions

While cGANs demonstrate remarkable potential for balancing minority classes, synthetic medical images carry severe clinical risks. Primary amongst these is high-frequency hallucination: the generator may synthesise plausible-looking pleural effusion or corneal opacities that lack physiological validity. Furthermore, Frechet Inception Distance (FID), calculated using ImageNet-trained Inception-V3 features, is suboptimal for grayscale veterinary radiographs where low-level thoracic vertebral counts matter more than natural scene textures. Future production pipelines should incorporate Physics-Informed Losses and Diffusion Models (DDPM) with clinician-in-the-loop validation.

7.0 References

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